

Pradeep Ponnamm

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EDUCATION

The University of Memphis

M.S. in Information Systems

Awards: 2024 Dean's List

May 2026

GPA: 4.0

TECHNICAL SKILLS:

Programming: Python, R, C, Java, SQL

AI/ML: PyTorch, TensorFlow, HuggingFace, YOLOv5, BERT, GPT-4, FAISS, GenAI, LLMs, Semantic Search

Tools/Cloud: AWS, GCP, Docker, GitHub Actions, TensorBoard, Firebase

Certifications: GenAI-LLMs, Machine Learning,

PROFESSIONAL EXPERIENCE:

The University of Memphis, Memphis, TN

Research Assistant

April 2025 – Present

- Created GPT-4 chatbot using FastAPI to help executives navigate US AI regulations, reducing onboarding time by 30%.
- Integrated BERT embeddings + geography profiles with LDA optimal topics, boosting user match accuracy by 25%.
- Designed regulatory training flows with tuned prompts, improving user comprehension scores by 20% in user tests.
- Applied retrieval augmented generation (RAG) and advanced prompt engineering + LLM fine-tuning, improving chatbot accuracy by 35% across 500+ queries.

ISRO, Hyderabad, India

Research Intern

January 2024 – April 2024

- Applied YOLOv5 to detect ships and perform instance segmentation of oil spills in SAR imagery, attaining 85% mAP.
- Optimized GPU-accelerated training + hyperparameter tuning, cutting inference latency by 40%.
- Created computer vision pipelines with data augmentation, boosting detection robustness by 25% across diverse ocean.
- Defined key performance metrics and benchmarked models using Python, TensorBoard and scikit-learn, enabling systematic evaluation and 20% precision gains Contributed to real-time satellite analytics for coastal surveillance.

BlackBuck Engineers, Hyderabad, India

Software Engineer Intern

May 2024 – August 2024

- Designed Spring Boot + Kafka microservices processing 10k+ daily transactions, cutting workflow delays by 35%.
- Delivered REST APIs serving 1000+ users, cutting response time by 25% and enabling real-time logistics coordination.
- Automated CI/CD with Docker + GitHub Actions on AWS EC2, reducing release cycles by 40% and 99% uptime.
- Integrated monitoring + distributed logging for microservices orchestration, reducing MTTR by 20%.

PROJECT EXPERIENCE:

Enterprise Document Processor – (SpringBoot, Kafka, OCR, BERT)

OCR Project Link

2025

- Built an enterprise document processing platform integrating OCR + BERT, automating entity extraction from PDFs.
- Engineered scalable microservices with Spring Boot + Kafka for distributed processing, handling 50k+ documents.
- Developed AWS S3 pipelines for secure storage and retrieval, cutting manual document handling by 60%.
- Designed integration with IAM STS role assumption and external IDs to manage infrastructure across environments.
- Implemented Docker, CI/CD pipelines, and monitoring dashboards, achieving 99% uptime in production workflows.

Drone Vision Object Detection – GLSAN, YOLOv5

2024

- Engineered aerial object detection system for vehicles + infrastructure in drone footage using Vis-Drone Dataset.
- Achieved +20% mAP over CNN baselines, improving recognition in low-light environments.
- Deployed optimized YOLOv5 models on GPU clusters, enabling near real-time aerial surveillance with 40% faster.

LEADERSHIP EXPERIENCE

CIS Tech Leadership Council, University of Memphis

Technical Program Lead

September 2024 – Present

- Led the development of technical workshops in Python, AWS, and SQL, focusing on business application use cases.
- Coordinated with faculty to create and deliver hands-on learning experiences, improving workshop attendance by 20%.